Group project I: from a dataset to a plan

Bioinformatic tools

Corentin Maslard

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Université de Neuchâtel
Faculty of Science



Quick poll: pe.app/cmaslard7

One question from Session 7 before we move on.

Since Session 3, you received the question, the data and the steps.

In Sessions 8 and 9, you decide.

Sessions 8 and 9: one project, four dates

Today, Mon 9 Nov

- Get your project
- Meet the data
- Write the plan and a first figure

PLAN.pdf by 23:59

Until Mon 16 Nov

- Each member writes one part of the analysis
- Meet at least once as a group

part_1 … part_5

Mon 16 Nov

- 12-min talk + 5 min of questions
- Everyone presents a part
- One question per member, about any part

Slides (PDF) by 15:15

Thu 19 Nov

- The host puts the parts together and renders the analysis

PROJECT.pdf + .qmd by 23:59

No new tool today: a new skill



Your toolbox: everything is already there

Shell, BLAST, alignments, trees

Sessions 2-3

Tables, plots, `lm()`

Session 4

PCA, heatmaps, clustering

Session 5

One model per feature, confounders

Session 6

Trees, forests, test sets

Session 7

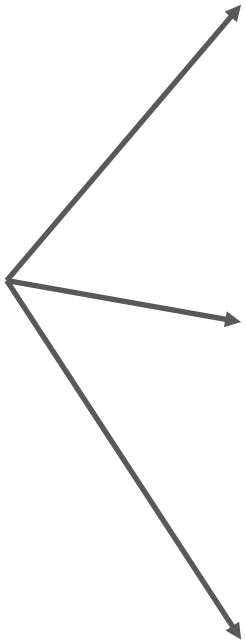
The new skill: turning **data** into a **question**, and a question into a **plan**.

1. From data to a question

One dataset, many questions

The soybean tables of Sessions 4-5

one row per sample and molecule, or per sample and bacterium



Which bacteria respond to drought?

→ one model per taxon

Do the two genotypes differ?

→ PCA, `lm()` with genotype

Does the root release molecules that the leaf does not contain?

→ compare the two organs

Same data, three questions, three different analyses.

A good question is …

Biological

About organisms, not about the data.

Which molecules change? is a step.

Does the plant make protective molecules in the root before the leaf? is a question.

Answerable with these data

The variables it needs are in the table.

Read `data/README.md` first.

Focused

One main question.

Two or three sub-questions at most.

Refutable

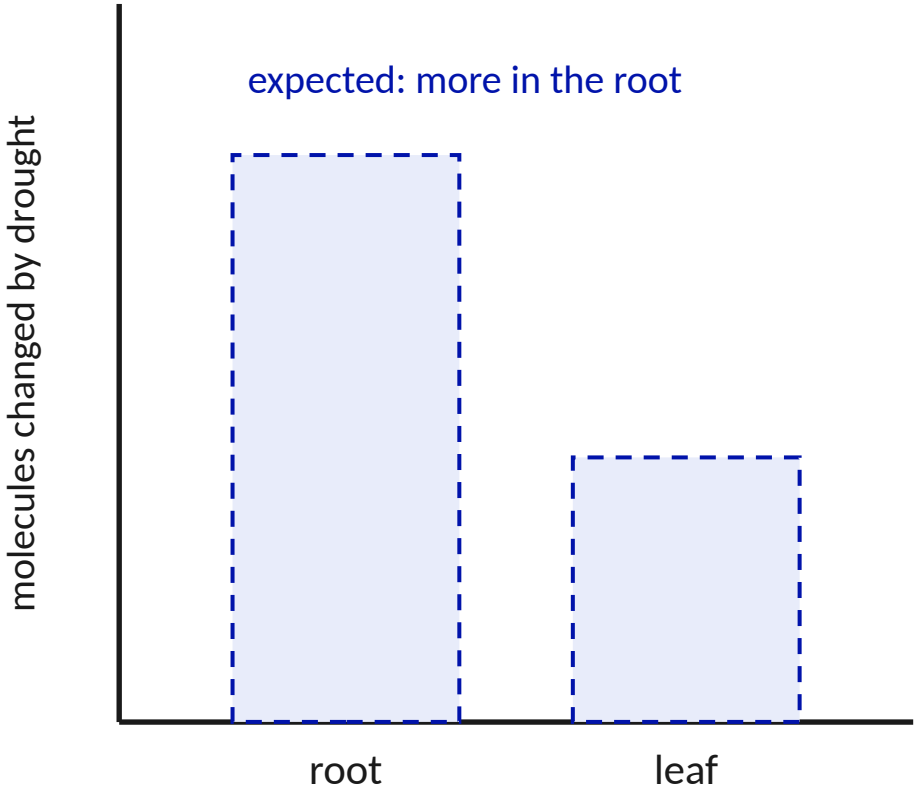
You can say in advance which result would mean **no**.

Draw the figure before the analysis

1. Write the question, in one sentence.
2. Draw the figure that would answer it: axes, colours, panels.
3. Write the result you expect on that figure.

Cannot draw the figure? The question is not precise enough yet.

Drawn before any code: soybean, drought



2. The anatomy of a plan

Seven short parts, about two pages and a figure.

The plan: seven short parts

Part	What to write	Example: the soybean plants of Sessions 4-5
Question	One sentence, then two or three sub-questions	Does drought change the chemistry of the root more than that of the leaf?
Hypothesis and refutation	What you expect, and the result that would prove you wrong	More molecules change in the root, which meets the dry soil first; refuted if the leaf has as many or more
Data	Files, rows, the subset you use, samples per group	One row per sample and molecule; root and leaf only; 2 genotypes × 4 conditions × 3-4 plants
Variables	Response, explanatory, possible confounders	Response: concentrations; explanatory: drought, organ; also in the model: genotype, heatwave
Method, step by step	Each step and the course tool that does it	1. log; 2. PCA per organ (S5); 3. one `lm()` per molecule with FDR (S5-6) …
First figure	One exploratory figure, its caption, what it already shows	PC1 vs PC2 of each organ, coloured by condition
Risks and roles	What could go wrong; who writes which part	Four plants per group is few; X writes the PCA part, Y the models …

3. The toolbox

Which tool for which question?

Which tool for which question?

If your question is …	Tool	Session
How many, how big, per group?	`group_by()`, `summarise()`, `count()`	4
Is there a difference between groups? Does it depend on a second factor?	`lm(y ~ a * b)`, `anova()`, boxplot	4
What are the main patterns among many variables? Which samples resemble each other?	`prcomp()`, heatmap, clustering	5
Which of thousands of features (genes, SNPs, OTUs) go with a variable?	one `lm()` per feature with `apply()`, Bonferroni or FDR	5, 6
Is an association real, or due to a confounder (ancestry, family, batch)?	the confounder in `lm()`, or PCs as covariates	6
Can a variable be **predicted** from the others, and how well on new data?	tree, random forest, test set, confusion matrix	7
Is this sequence similar to a known one? How are these genes related?	`blastp`, `mafft`, `FastTree`	3
Where is it?	a map with `geom_point()` or `geom_tile()`	4, 6

Rule 1

First explore, then test

A plot of the raw data comes before any model.

It shows the outliers, the batches, the missing combinations.

Rule 2: one question, several tools

1. Explore: what is one row, how many per group, one raw plot
2. PCA: the main patterns (Session 5)
3. One model per gene, corrected for thousands of tests (Sessions 5-6)
4. A closer look at the best genes: what are they?

A good project chains three or four tools.

4. The classic traps

You met each of them in the course.

The traps come back in real data

The confounder

Session 6

A third variable creates the association: northern accessions flower late *and* share alleles.

The batch

Day, person, experiment. Does it explain your first PC?

Thousands of tests

Sessions 5-6

20,000 genes: 1,000 "significant" at 5% by chance alone.

Unequal sampling

Ten times more records in 2020: first, ten times more observers.

Raw counts

Twice the reads, twice the counts. Normalise: per million, then log.

Overfitting

Session 7

Judged on its own training data, a model always looks good. Keep a test set.

Correlation is not causation

Observations: say what an experiment would need to show.

Your project

Read its *Watch out*: which of these traps are in your data?

What we look at

Today: the plan (group)

- A focused biological question
- The variables and the expected result, and what would refute it
- The method step by step, each step with a tool of the course
- A first figure with its caption
- One difficulty or decision

Session 9: the talk

The group

- Question and context; a correct, reproducible analysis; clear figures; conclusions supported by the results; one limitation

You

- Your part of the talk: clear, and you master it
- Your answer to one question, about any part of the project

The seven projects

Two minutes each. Then each group ranks three.

Seven datasets, seven biologies

	Project	Organisms	Data
A	How fast does a plant answer a bacterium?	*Arabidopsis*, *Pseudomonas*	RNA-seq, 366 samples, time series × mutants
B	What controls the expression of a gene in nature?	*Arabidopsis*, 649 wild accessions	RNA-seq + the SNPs of Session 6
C	When does a coral break down under heat?	*Stylophora* coral and its algae	RNA-seq of both partners, 80 samples
D	What shapes the communities of ocean bacteria?	Marine bacteria and archaea	16S counts + ocean chemistry, 139 samples
E	What does the shape of a bird tell about its life?	11,009 bird species	Morphology, ecology, geography
F	How many immune genes does a plant need?	413 plant genomes	Immune receptor genes per genome + sequences
G	Will species move with the climate?	8 European species	GBIF observations + climate now and in 2070

A. How fast does a plant answer a bacterium?

The biology

- Bacteria inject **effectors**; plant **R proteins** recognise some of them and trigger a stronger, faster response.
- *Pseudomonas* DC3000 is virulent: nothing is recognised. With **AvrRpt2** (seen by RPS2) or **AvrRpm1** (seen by RPM1), the plant wins.
- Hormones carry the signal: salicylic acid (*SID2*), *PAD4*, *EIN2* (ethylene), *DDE2* (jasmonate).

The data

- Leaf RNA-seq: 366 samples, 18,422 genes
- 6 genotypes (wild type, hormone and receptor mutants) × mock or 3 strains × 1 to 24 h, 3 trials

Starting question

Does wild-type *Arabidopsis* respond faster to a bacterium it recognises than to one it does not? How many genes change, and when?

Tools

Normalisation; PCA; infected vs mock per time point, one model per gene with a correction; heatmap (Sessions 4-6)

Folder `A_arabidopsis_pseudomonas` · Mine et al. 2018, *The Plant Cell*

B. What controls the expression of a gene in nature?

The biology

- Session 6: the genes behind flowering time, in 1,003 wild accessions. Here the trait is the **expression of a gene**.
- A variant next to the gene (*cis*), or in another gene that regulates it (*trans*)?
- A GWAS of an expression level is an **eQTL** scan.

The data

- Leaf RNA-seq of 649 accessions, 20,832 genes, one growth chamber
- Their SNPs (Session 6), flowering time, climate of origin

Tools

`lm()`, correlation; PCA; GWAS with `apply()`, QQ plot, PCs as covariates; test sets (Sessions 4-7)

Starting question

FLC blocks flowering. Is its expression associated with flowering time? Which regions of the genome control it: *FLC* itself, or others?

Folder `B_arabidopsis_1001_transcriptomes` · Kawakatsu et al. 2016, *Cell*; Session 6

C. When does a coral break down under heat?

The biology

- A coral is a partnership: the animal hosts algae that feed it by photosynthesis.
- Too warm: the coral loses its algae and turns white (**bleaching**).
- Corals of the northern Red Sea are unusually heat-tolerant.

The data

- RNA-seq of both partners, coral and algae: 80 samples
- 5 colonies × 27 to 34.5 °C × short or long heat × end of heat or recovery

Tools

Normalisation; PCA; `lm()` with several factors, share of variation; one model per gene (Sessions 4-6)

Starting question

At which temperature does the coral switch on a large stress response, and does it recover? Does it differ between the short and the long experiment?

Folder `C_coral_heat_stress` · Savary et al. 2021, *PNAS*

D. What shapes the communities of ocean bacteria?

The biology

- Ocean bacteria and archaea make half of the photosynthesis on Earth.
- The schooner *Tara* sampled the oceans at three depths: surface, deep chlorophyll maximum, dark mesopelagic (200-1,000 m).
- 16S reads grouped into taxa (OTUs), as the soybean microbiome of Session 4.

The data

- 139 samples: taxa (OTU counts) and functions (KEGG modules)
- Depth, temperature, oxygen, nutrients, ocean

Tools

Relative abundances, Shannon diversity, `lm()`; PCA of the communities; maps (Sessions 4-6)

Starting question

Which factor best explains the composition of the bacterial community: depth, temperature, or the ocean?

Folder `D_tara_oceans_plankton` · Sunagawa et al. 2015, *Science*

E. What does the shape of a bird tell about its life?

The biology

- The beak follows the diet, the wing the distance flown, the legs where the bird moves.
- AVONET: about 90,000 birds measured, for 11,009 species.
- Ecological rules can be tested, such as **Bergmann's rule**: larger in colder climates.

The data

- One row per species: beak, wing, tail, legs, mass; diet, habitat, migration, range
- One row per measured bird

Tools

Log of the measurements; PCA; `lm()` with a covariate; trees, forests, test set (Sessions 4-7)

Starting question

Can you tell what a bird eats from the shape of its body? Which measurements separate the diets best?

Folder `E_avonet_birds` · Tobias et al. 2022, *Ecology Letters*

F. How many immune genes does a plant need?

The biology

- **NLRs**: the main immune receptors inside plant cells (the R genes of project A).
- Three classes: **TNL**, **CNL**, **RNL** (helpers). TNLs need **EDS1** to signal.
- *Arabidopsis*: about 170 NLR genes; bread wheat more than 2,000; some plants almost none.

The data

- 413 plant genomes (UniProt), from algae to trees: the domains of each protein
- One row per NLR gene; the NLR proteins of 8 species, for BLAST and trees

Tools

Tables, `lm()` with a covariate; `blastp`, `mafft`, `FastTree`; confounders (Sessions 3-6)

Starting question

How many NLR genes do plants have? Does this number follow their lineage, the size of their genome, or their way of life?

Folder `F_plant_immune_genes` · UniProt 2025; Baggs et al. 2020, *The Plant Cell*

G. Will species move with the climate?

The biology

- Warming should push species towards the poles and up the mountains.
- **GBIF** gathers millions of observations; with climate maps, they describe the **climatic niche** of a species.
- A model of presence from climate can be projected onto 2061-2080.

The data

- 8 European species (4 warm-loving, 2 cold-adapted, 2 trees), 1970-1999 and 2000-2024
- 38,004 grid cells: climate now and in 2061-2080 (two scenarios)

Tools

Maps with `geom_tile()`; `lm()`; trees, random forest, test set (Sessions 4 and 7)

Starting question

In which part of the European climate does each species live? Has the range of the warm-loving species moved north?

Folder `G_species_ranges_climate` · GBIF; WorldClim 2 (Fick and Hijmans 2017)

Choose your project

1. The host registers the group on Moodle, **Group project: registration**: the five names and three projects in order. Until 15:00.

2. At 15:00, the groups are taken in an order drawn by lot. Each receives the first project of its list that is still free.

3. A group whose three projects are taken chooses among the free ones, in the drawn order.

Registering early gives no advantage. One project stays unused.

Setup: every member, in the Terminal

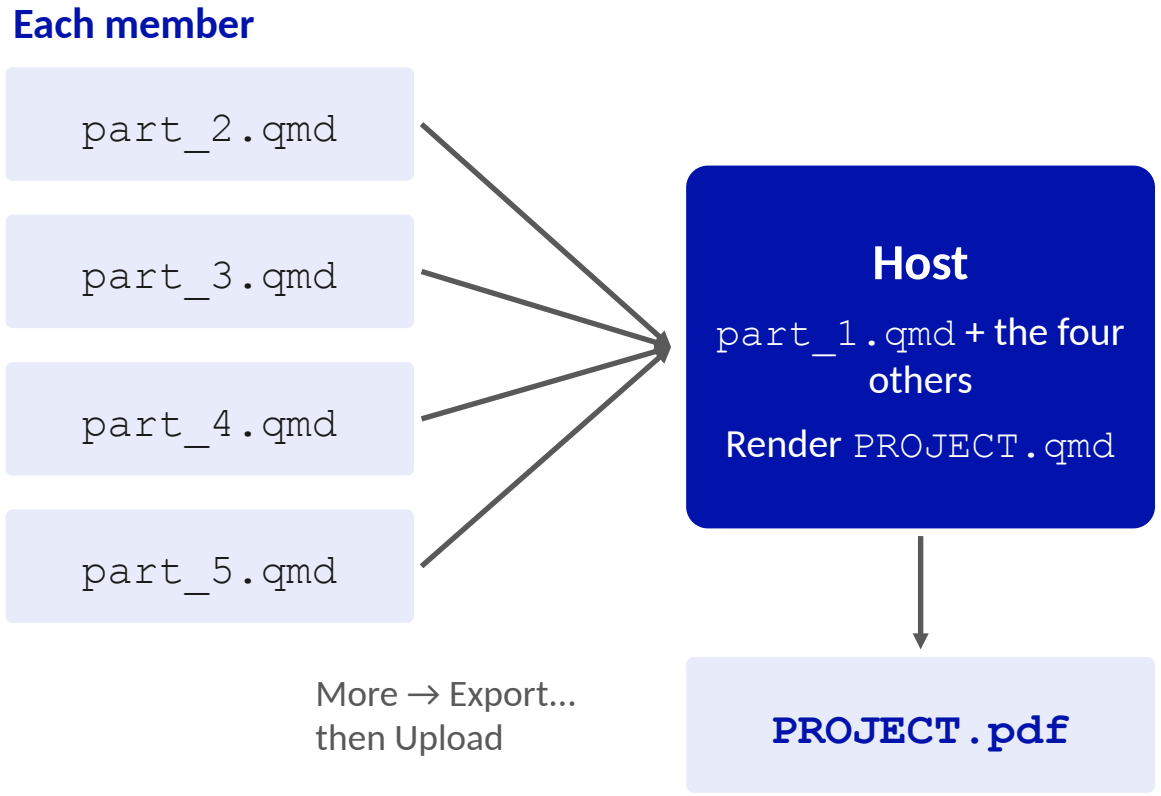
```
PROJECT=C_coral_heat_stress     # your project's folder
mkdir -p ~/bioinfo_course/session_08
cp -r --update=none /home/cmaslard/bioinfo_tools_course/session_08/$PROJECT \
      ~/bioinfo_course/session_08/
ls -lh ~/bioinfo_course/session_08/$PROJECT
```

- Then **File → New Project → Existing Directory**: your project folder.
- Open `data/README.md` and read it all: what one row is, what every column means.
- Run the section **The data** of `PROJECT.qmd`: it reads every table.
- Choose the **host**: the member who keeps `PLAN.qmd` and `PROJECT.qmd`, collects the parts and submits.

One folder each, one part each

```
C_coral_heat_stress/
  data/           tables + README.md
  PLAN.qmd        today's plan
  PROJECT.qmd     The data + the parts
  part_1.qmd      one part per member
  ...
  part_5.qmd
  results/        figures (ggsave)
```

- **The data**: written together, the same in every copy.
- A part starts from **The data** only: it runs alone.
- No labels on the chunks of the part files.



Render `PROJECT.qmd` in your own copy to check your part: the other parts are empty there.

Today, from 15:05



A Setup, then meet the data: what is one row? How many samples per group? Missing values? One raw plot.

B The plan in `PLAN.qmd`: question, hypothesis, variables, method, first figure, risks, roles.

C Three minutes with me: your question in one sentence, your first figure, your first two steps.

D Who writes which part. Render `PLAN.pdf`: the host submits it by **23:59**.

Turn data into a question, and a question into a plan.

Everything else, you already know.